Certified Bounds for Partially Coherent Mask Synthesis: Convex Printability Certificates, Certified Separation of Single- and Multi-Exposure Optima, and Pattern-Specific Exposure-Latitude Limits

Autonomous research system (Claude Code) with human oversight*

October 12, 2026

Abstract

Purpose. Mask synthesis for projection lithography (inverse lithography, ILT; source–mask optimization, SMO) is nonconvex, so a local optimum says nothing about the best achievable image. We provide solver-independent, checkable limits on printability.

Methods. For the Abbe/Hopkins model of a periodic cell—scalar or vector, with arbitrary aberrations and focus sets—the best achievable dark-to-bright intensity ratio $\rho^\star$ (equivalently the maximum full-range exposure latitude of a threshold resist at a prescribed edge tolerance) is bounded by a semidefinite relaxation in the diffraction-order domain. Its dual is a certificate: nonnegative weights on dark and bright sample points such that one explicit matrix is positive semidefinite, verified independently of the solver by rebuilding the certificate matrix in 50-digit arithmetic from the model equations and factorizing it with a positive margin. We prove that the relaxation equals the unlimited-multiple-exposure problem, that the collected spectra of nonnegative one-dimensional masks form the Carathéodory–Toeplitz cone, that complex transmission cannot improve the relaxation under symmetric illumination, and that pixelation at or above the collected-order count leaves the bound unchanged. A branch-and-bound certifier (interval bounds of positive semidefinite forms, and an SDP-bounded variant with verified dual certificates) bounds the single-mask optimum.

Results. The bound reproduces the classical k_1 cutoffs for DUV and EUV and the two-beam closed form for dense lines. On 108 non-degenerate random one-dimensional targets (one near-degenerate instance is reported separately), local optimization is certified optimal to within 10^{-4} in 74.1% of cases and to within 1% in 85.2%; all two-dimensional contact arrays are certified within 2.25%, and the annular T-junction and line-end cells within 0.20% and 0.04%. Strict multi-exposure gains of 6.4%–16.4% (random coherent targets, converged branch-and-bound) and 8.9% (mixed-width target) are certified against the ratio realized by a feasible multi-exposure ensemble. The bound is 1.5–135× tighter than the Rayleigh-quotient bound, brackets the penalty of pixelated nonnegative masks relative to phase-shift masks between certified lower bounds of 2.6%–11.4% and upper bounds of 3.6%–17.6%, and brackets the suboptimality of a standard pattern-error ILT at 19%–32% above the phase-shift optimum.

Conclusions. The certificates make local optimality checkable, quantify the multi-exposure and phase-shift entitlement, and retain their structure under the vector model and moderate aberrations; their numerical values are entitlements of the thin-mask aerial-image model, necessary but not sufficient for printability.

Keywords: computational lithography; inverse lithography; source–mask optimization; partially coherent imaging; semidefinite relaxation; duality certificates; branch and bound; exposure latitude.

*Correspondence: repository owner. Code, data and the full verification record accompany this manuscript (`code/`, `results/`, `notes/`). Every number and every table body is generated from the result files of one clean execution of `code/run_all_par.sh` by `code/make_numbers.py`; no number is typed by hand. Manuscript v3: internally verified and revised after six independent adversarial reviews; not peer reviewed.

1 Introduction

Computational lithography synthesizes a mask (and a source) so that the projected aerial image prints a target layout. In the Abbe formulation the image is a weighted sum, over source points, of squared moduli of filtered mask spectra: it is *quadratic* in the mask. ILT and SMO therefore optimize nonconvex functionals. IBM's wavefront-optimization work states that mask optimization "tends inherently to be a problem with non-convex quadratic constraints" [1]; recent work on learned ILT describes objectives "that often trap optimization in poor local minima" [2]; and reviews list local optimality among the open problems [3, 4]. What is missing is a *certificate*: a proof, for a given target, illumination, numerical aperture and wavelength, that no mask can do better than a stated value. Such certificates (i) separate unprintable layouts from unconverged optimizers, (ii) benchmark ILT engines against the true optimum, (iii) quantify the value of multiple exposure and of phase-shift versus nonnegative masks, (iv) generalize the k_1 rule to arbitrary layouts, and (v) localize the geometry that limits printability through the support of the dual weights.

Contributions.

(C1) Theorem 1: dual certificates of printability, valid for any mask class and for incoherent sums of masks, verified by one PSD test independent of the solver, re-derived exactly in 50-digit arithmetic from the model equations (§5).

(C2) Theorem 3: the relaxation equals the unlimited-multi-exposure problem, and Pataki's bound limits the number of useful exposures. Lemma 7: complex transmission cannot improve the relaxation under symmetric illumination. Theorem 5: Carathéodory–Toeplitz realizability for nonnegative one-dimensional masks. Lemmas 8–9: vector imaging, aberrations and pixelation preserve the structure.

(C3) Theorem 10 and §5: branch-and-bound certificates of the single-mask optimum which, combined with (C1), separate single- from multi-exposure optima on concrete targets.

(C4) An open pipeline, an empirical study whose every number is generated from the result files, and independent verification of every certificate.

Not claimed. No new SDP algorithm (the Shor relaxation of a QCQP is textbook [5, 6], and SDP-bounded branch-and-bound for nonconvex QCQPs is established [7]); no full-chip scalability; no thick-mask electromagnetics, resist chemistry or stochastics; no statement about proprietary ILT engines. "Certified" means exact in exact arithmetic and safeguarded, not interval-verified, against double-precision rounding (§5).

Organization. §3 states the imaging model, the synthesis problem and its relaxation; §4 proves the certificate, the multi-exposure identity, the Toeplitz characterization and the branch-and-bound bound; §5 describes the solver, the round-off safeguard and the primal heuristics; §6 reports the experiments; the Discussion and Limitations close the paper.

2 Related Work

Wavefront optimization. Rosenbluth et al. [1, 8] optimize the collected diffraction orders directly, maximizing exposure latitude subject to sampled bright/dark intensity constraints—the same objective as ours; a patent of the same group writes the problem as a quadratic program in the order amplitudes [9], and the source step is a linear program [10]. Their global strategies partition the solution space into regions with at most one local minimum [1], use

dominant "joint eigenvectors" and a "step-walk relaxation", and, for multiple exposure, an unconstrained Rayleigh-quotient simplification [9]. None of these provides a certificate for the pointwise max-dark/min-bright ratio of a solution obtained by other means: the partition and step-walk strategies are search heuristics, and the multiple-exposure patent minimizes an *averaged* dark-to-bright objective whose global minimum would bound the pointwise ratio only through that weaker averaged quantity; we compare with its Rayleigh-quotient bound in Table 8. The stacked pupil-shift matrix of [11] is the channel matrix of our Eq. (2); pixel-based ILT [12, 13] and SMO [14] are the local methods our certificates evaluate.

Quadratic programs and SDP in lithography. Nonaka et al. [15] write pixel mask optimization as a 0–1 convex QCQP and relax it heuristically (no bound); Nouralishahi, Wu and Vandenberghe [16] use semidefinite programming for lithography *model calibration* with a PSD imaging-kernel constraint, not for synthesis; PSD relaxations with randomized rounding serve the combinatorial mask-assignment problem of multiple patterning [17]. Liu and Zakhor [18, 19] designed binary and phase-shift masks by branch and bound on a pixel least-squares criterion (combinatorial).

SDP relaxation of QCQPs. The Shor relaxation and its approximation theory are surveyed in [5]; max-min multicast beamforming [20] has exactly the structure of coherent-illumination mask synthesis. Pataki [21] bounds the rank of extreme solutions; Lasserre's hierarchy [22] tightens bounds for small problems; SDP-bounded branch-and-bound solves nonconvex QPs globally to a tolerance [7]. The Toeplitz-positivity description of nonnegative trigonometric sequences is standard in filter design [23, 24].

Bounds in photonic design. Angeris, Vučković and Boyd [25, 26] bound inverse-designed photonic devices by Lagrangian duality; our work is the imaging-side analogue.

Information-theoretic limits and lifting. Information-theoretic limits [27, 28] bound OPC fidelity through mutual information; these are not layout-specific duality certificates. PhaseLift [29] lifts quadratic measurements for recovery; we lift for synthesis. The optimal coherent decomposition of partially coherent imaging is due to [30].

3 Model and Problem Statement

3.1 Imaging model

Let $\Lambda = [0, L_1) \times \cdots \times [0, L_d)$, $d \in \{1, 2\}$, be a periodic cell and $m : \Lambda \to \mathbb{C}$ the mask transmission with Fourier coefficients $M_k = |\Lambda|^{-1} \int_\Lambda m(x) e^{-2\pi i f_k \cdot x}\, dx$, $f_k = (k_1/L_1, k_2/L_2)$ (for $d = 1$, $f_k = (k/L, 0)$: a one-dimensional mask is invariant along y, but the illumination is still two-dimensional). Illumination is a finite area-weighted quadrature of the illuminator: source points $s_j \in \mathbb{R}^2$ (direction cosines, $|s_j| \le \sigma\,\mathrm{NA}$) with weights $w_j > 0$, $\sum_j w_j = 1$; the illuminators used are listed in §6. In an image-side medium of index n the pupil is

$$P_z(g) = \mathbb{1}\{|g|\lambda \le \mathrm{NA}\}\, \exp\Big(i\tfrac{2\pi n z}{\lambda}\big(\sqrt{1 - |g|^2\lambda^2/n^2} - 1\big) + iW(g)\Big), \tag{1}$$

with defocus z and aberration phase W. Only the collected orders $K = \{k : |f_k| \le (\mathrm{NA} + \max_j |s_j|)/\lambda\}$ contribute; an order k is discarded from K only if neither k nor $-k$ is brought into the pupil by any source point, so that $K = -K$ (needed for the real-mask structure of Lemma 2; a dead order whose mirror is collected stays in K as an unconstrained coordinate

tied to its mirror by $M_{-k} = \overline{M_k}$), and $F = |K|$; $M \in \mathbb{C}^F$ denotes $(M_k)_{k\in K}$. The Abbe image at x and focus z is

$$I_{M,z}(x) = \sum_j w_j \Big| \sum_{k\in K} P_z(f_k + s_j/\lambda)\, M_k\, e^{2\pi i f_k\cdot x} \Big|^2 = M^{\mathsf{H}} Q_{x,z} M, \qquad Q_{x,z} = \sum_j w_j c_{j,x,z} c_{j,x,z}^{\mathsf{H}} \succeq 0. \tag{2}$$

This is the scalar thin-mask model; the vector extension is Lemma 8.

3.2 Synthesis problem and relaxation

A target is specified by finite sample sets B (bright) and D (dark), placed at $\pm\tau$ (edge tolerance) from every target edge plus interior points, and by a focus set Z; throughout, τ is 10% of the smallest critical dimension of the target (one-dimensional samples sit exactly at $\pm\tau$; two-dimensional targets are unions of axis-aligned rectangles and their samples lie exactly on the two offset contours at distance τ from the boundary of the union, every 2 nm along the contour including the circular arcs around convex corners; the exact distance of every sample is asserted to be at least τ). Mask classes: $\mathcal{M}_{\mathbb{C}} = \mathbb{C}^F$; real masks $\mathcal{M}_{\mathbb{R}} = \{M : M_{-k} = \overline{M_k}\}$ (continuous-tone transmission in $[-1, 1]$, the class containing phase-shift masks); nonnegative masks $\mathcal{M}_+ \subset \mathcal{M}_{\mathbb{R}}$ (binary-like masks; an attenuated phase-shift mask with negative transmission belongs to $\mathcal{M}_{\mathbb{R}}$). The dark-to-bright ratio

$$\rho(M) = \frac{\max_{(x,z)\in D\times Z} I_{M,z}(x)}{\min_{(x,z)\in B\times Z} I_{M,z}(x)} \qquad (\rho := +\infty \text{ if the denominator vanishes}) \tag{3}$$

is scale-invariant. For a threshold resist (threshold θ, dose E) every bright sample prints and no dark sample prints iff $\theta/\beta \le E < \theta/\delta$ with $\beta = \min_{B\times Z} I$, $\delta = \max_{D\times Z} I$; the full-range dose latitude, window width over its centre dose, is

$$\mathrm{EL}(\rho) = \frac{2(1-\rho)}{1+\rho}, \tag{4}$$

with one dose over the whole focus set (the half-range value $(1-\rho)/(1+\rho)$ equals the Michelson contrast of the two extreme samples). The synthesis problem and its relaxation are

$$\begin{aligned} (\mathrm{P}_{\mathcal{M}}) \quad \rho^\star(\mathcal{M}) &= \inf_{M\in\mathcal{M}\setminus\{0\}} \rho(M) \\ &= \inf\big\{t :\ M^{\mathsf{H}} Q_d M \le t \ \ \forall d \in D\times Z, \ \ M^{\mathsf{H}} Q_b M \ge 1 \ \ \forall b \in B\times Z\big\}, \qquad (5) \\ (\mathrm{R}) \quad t_R &= \min\big\{t :\ \mathrm{Tr}(Q_d X) \le t \ \ \forall d \in D\times Z, \ \ \mathrm{Tr}(Q_b X) \ge 1 \ \ \forall b \in B\times Z, \\ &\qquad X \succeq 0 \ \ [\, X = \mathrm{conj}(\Pi X \Pi)\,]\big\}, \qquad (6) \end{aligned}$$

where Π is the order-flip permutation and the bracketed constraint selects real masks.

3.3 Notation

ρ is the ratio of a given mask, $\rho^\star$ its infimum over a class, t_R the value of (R); "bound" always denotes the rigorized certificate value (a lower bound on $\rho^\star$, hence an upper bound on the exposure latitude), "best single mask" the best feasible real mask found by local search, and "gap" = (best single mask − bound)/bound. Subscripted λ_b is a dual weight; unsubscripted λ is the wavelength. Table 1 collects the remaining symbols.

4 Theory

Theorem 1 (Certificate)**.** *Let $\mu \ge 0$ on $D\times Z$ with $\sum \mu = 1$, $\lambda \ge 0$ on $B\times Z$, and $S(\mu,\lambda) = \sum_d \mu_d Q_d - \sum_b \lambda_b Q_b \succeq 0$. Then $\rho(M) \ge \sum_b \lambda_b$ for every $M \in \mathbb{C}^F \setminus \{0\}$ and for every incoherent sum of images $\sum_i I_{M_i}$. For real masks it suffices that $\tilde{S} = \mathrm{Re}(U^{\mathsf{H}} S U) \succeq 0$ with U a unitary basis of $\mathcal{M}_{\mathbb{R}}$.*

Table 1: Notation. Intensities are in arbitrary units (ρ is dimensionless); lengths in nm unless stated.

Symbol	Meaning
Λ, L_i, d	periodic cell, its side lengths, and its dimension ($d \in \{1, 2\}$)
m, M_k, M	mask transmission, its Fourier coefficients, and the vector of collected coefficients
K, $F = \lvert K\rvert$	set and number of collected diffraction orders
λ, NA, σ, n	wavelength, numerical aperture, partial-coherence factor, image-side refractive index
s_j, w_j	source points (two-dimensional direction cosines) and their quadrature weights
P_z, z, W, Z	pupil function, defocus, aberration phase, finite focus set
$I_{M,z}(x)$, $Q_{x,z}$	aerial image at (x, z) and its Hermitian PSD form in the orders
B, D, τ	bright and dark sample sets; edge tolerance (10% of the minimum CD)
$\mathcal{M}_{\mathbb{C}}$, $\mathcal{M}_{\mathbb{R}}$, $\mathcal{M}_{+}$	complex, real and nonnegative mask classes
ρ, $\rho^\star$, t_R, EL	dark-to-bright ratio, its optimum over a class, value of (R), full-range exposure latitude (4)
μ_d, λ_b, S, $\tilde{S}$	dual weights; certificate matrix $\sum_d \mu_d Q_d - \sum_b \lambda_b Q_b$ and its real-structured form
Π, J, U, X	order-flip permutation, antilinear involution $\mathrm{conj}(\Pi\,\cdot)$, unitary basis of $\mathcal{M}_{\mathbb{R}}$, lifted PSD variable
κ	rigorization scale of a certificate (§5)
L, U (Theorem 10)	certified single-mask lower bound and incumbent upper bound
k_1	normalized half-pitch, $k_1 = \text{half-pitch} \cdot \mathrm{NA}/\lambda$

Proof. Let $\beta = \min_{B\times Z} I > 0$ and $\delta = \max_{D\times Z} I$. Then $\delta = \delta \sum_d \mu_d \geq \sum_d \mu_d M^{\mathsf{H}} Q_d M = M^{\mathsf{H}} S M + \sum_b \lambda_b M^{\mathsf{H}} Q_b M \geq \beta \sum_b \lambda_b$. For sums of images apply the same chain termwise. For $M = Ur$, $r \in \mathbb{R}^F$, $M^{\mathsf{H}} S M = r^{\mathsf{T}} \tilde{S} r$. □

Lemma 2 (Real structure)**.** *$J(M) = \mathrm{conj}(\Pi M)$ is an antilinear involution; $\mathcal{M}_{\mathbb{R}} = \mathrm{Fix}(J)$ is a real F-dimensional subspace on which the Hermitian form is real. A PSD X satisfies $X = \mathrm{conj}(\Pi X \Pi)$ iff $XJ = JX$; then $X = UYU^{\mathsf{H}}$ with $Y \in \mathbb{S}_+^F$ and $\mathrm{Tr}(QX) = \mathrm{Tr}(\tilde{Q}Y)$, $\tilde{Q} = \mathrm{Re}(U^{\mathsf{H}} Q U)$. Hence (R) for real masks is a real SDP of the same size.*

Theorem 3 (Relaxation equals unlimited multiple exposure)**.** *For $\mathcal{M} \in \{\mathcal{M}_{\mathbb{C}}, \mathcal{M}_{\mathbb{R}}\}$, t_R equals the optimal ratio of the summed image of any number of incoherently superposed exposures of masks in $\mathcal{M}$ printed with one common threshold. Assume every bright form is nonzero ($Q_b \neq 0$ for all b; otherwise $\rho^\star = t_R = +\infty$). Then (R) attains its minimum and $\rho^\star(\mathcal{M})$ is attained by a mask (Lemma 4); $\rho^\star = t_R$ iff (R) has a rank-one optimal solution; and an extreme optimal solution of (R) has $\mathrm{rank}\, X = r$ with $r(r+1)/2 \leq \lvert D \times Z\rvert + \lvert B \times Z\rvert$ for $\mathcal{M}_{\mathbb{R}}$ and $r^2 \leq \lvert D \times Z\rvert + \lvert B \times Z\rvert$ for $\mathcal{M}_{\mathbb{C}}$, so at most r exposures are needed.*

Proof. $X \succeq 0 \iff X = \sum_{i \leq F} M_i M_i^{\mathsf{H}}$, and $\mathrm{Tr}(QX) = \sum_i M_i^{\mathsf{H}} Q M_i$ is the summed image; Lemma 2 gives real M_i in the real-structured case. Attainment is Lemma 4. If $\rho^\star = t_R$, an optimal mask (which exists by the lemma) gives a rank-one optimal X; conversely a rank-one optimal $X = MM^{\mathsf{H}}$ is an optimal mask. The optimal face of (R) is a face of a feasible set that contains no line, so it has extreme points, which are extreme points of the feasible set. Writing the inequalities with slacks, (R) is a standard-form SDP with $m = \lvert D\times Z\rvert + \lvert B\times Z\rvert$ equalities and block-diagonal cone $(X, \text{slacks}, t^+, t^-)$; Pataki's theorem [21] bounds $\sum_{\text{blocks}} r_b(r_b+1)/2 \leq m$ for real symmetric blocks and $\sum_b r_b^2 \leq m$ for Hermitian blocks. □

Lemma 4 (Attainment)**.** *Let $\mathcal{N} = \bigcap_d \ker Q_d$ be the common dark null space, P the orthogonal projector onto $\mathcal{N}^\perp$, $B_0 = \{b : Q_b = PQ_bP\}$ the bright forms vanishing on $\mathcal{N}$ and $B_{\mathcal{N}} = B \setminus B_0$. Assume $Q_b \neq 0$ for all b. Then (R) attains its minimum, and $\rho^\star(\mathcal{M})$ is attained, for $\mathcal{M} \in \{\mathcal{M}_{\mathbb{C}}, \mathcal{M}_{\mathbb{R}}\}$.*

Proof. Since $Q_d = PQ_dP$, every dark value depends on X only through PXP, and so does every bright value with $b \in B_0$. (i) *Reduction.* Let (R_0) be (R) restricted to $A \succeq 0$ supported on $\mathcal{N}^\perp$ with the bright constraints of B_0 only. If X is feasible for (R) then $A = PXP$ is feasible for (R_0) with the same t. If A is feasible for (R_0) then $X_c = A + c(I-P) \succeq 0$ has the same dark and B_0 values and $\mathrm{Tr}(Q_b X_c) = \mathrm{Tr}(Q_b A) + c\,\mathrm{Tr}((I-P)Q_b(I-P))$ with $\mathrm{Tr}((I-P)Q_b(I-P)) > 0$ for $b \in B_{\mathcal{N}}$, so X_c is feasible for (R) for c large. Hence both problems have the same value and attain together. (ii) (R_0) *attains.* Its dual is $\max \sum_{b\in B_0} \lambda_b$ subject to $\sum_d \mu_d = 1$, $\mu, \lambda \geq 0$ and $\sum_d \mu_d Q_d - \sum_{B_0} \lambda_b Q_b \succeq 0$ on $\mathcal{N}^\perp$. By definition of $\mathcal{N}$, $\sum_d Q_d$ is positive definite on $\mathcal{N}^\perp$, so uniform μ and small positive λ are strictly dual feasible; (R_0) is feasible ($A = cP$ with c large, since $\mathrm{Tr}(Q_b P) = \mathrm{Tr}(Q_b) > 0$ for $b \in B_0$); strict dual feasibility of a feasible SDP implies that the primal optimum is attained [31]. (iii) *Single masks.* Write $M = u + v$ with $u \in \mathcal{N}^\perp$, $v \in \mathcal{N}$. Dark values and B_0 values depend only on u. For v outside the finitely many proper subspaces $\ker((I-P)Q_b(I-P)) \cap \mathcal{N}$, $b \in B_{\mathcal{N}}$, and $c \to \infty$, every bright value with $b \in B_{\mathcal{N}}$ of $u + cv$ tends to $+\infty$. If $B_0 = \emptyset$, $M = v$ has zero dark and positive bright values, so $\rho^\star = 0$ is attained. Otherwise $\rho(u+v) \geq \rho_0(u) := \max_d Q_d[u] / \min_{B_0} Q_b[u]$ for every v, with equality for v generic and c large, so $\rho^\star = \inf_{u \neq 0} \rho_0(u)$. On the unit sphere of $\mathcal{N}^\perp$ the numerator of ρ_0 is continuous and positive and the denominator continuous and nonnegative, so ρ_0 is a lower semicontinuous map into $(0, +\infty]$ on a compact set and attains its infimum at some $u^\star$; then $M = u^\star + cv$ attains $\rho^\star$. For $\mathcal{M}_{\mathbb{R}}$ the same argument applies to the real forms $\tilde{Q}$ of Lemma 2. (If $\mathcal{N} = \{0\}$ the sphere argument applies directly.) □

Remark. Theorem 3 models double or multiple exposure of one resist with a common development threshold; Boolean multi-patterning (litho–etch–litho–etch) is a different process and is not bounded by t_R. A gap between a locally optimized single mask and t_R is an upper bound on the multi-exposure benefit; it proves a strict benefit only when a certified single-mask lower bound exceeds the ratio of a feasible multi-exposure ensemble (§5, Theorem 10).

Theorem 5 (Nonnegative masks in one dimension)**.** *Let $d = 1$, $K = \{-K_{\max}, \dots, K_{\max}\}$, $T(M) = (M_{a-b})_{a,b=0}^{K_{\max}}$ the Hermitian Toeplitz matrix of $M \in \mathcal{M}_{\mathbb{R}}$, and let $\mathcal{C}_+$ denote the set of in-band coefficient vectors of bounded nonnegative masks. Then*

$$\{M : T(M) \succ 0\} \subset \mathcal{C}_+ \subset \{M : T(M) \succeq 0\} = \{\textit{in-band coefficients of nonnegative measures}\},$$

both inclusions are strict, $\overline{\mathcal{C}_+} = \{M : T(M) \succeq 0\}$, and the nonnegative-mask problem $\inf_{M \in \mathcal{C}_+} \rho(M)$ has the same value as its relaxation over $\{T(M) \succeq 0\}$ (the latter is an outer approximation; the former need not attain the value). If $T(M) \succ 0$, M is realized by a bounded, continuous, strictly positive mask.

Proof. Necessity: $v^{\mathsf{H}} T(M) v = |\Lambda|^{-1} \int m(x) |\sum_a v_a e^{2\pi i a x/L}|^2 dx \geq 0$ for a nonnegative function or measure. The equality on the right is the Carathéodory–Toeplitz theorem [24, 32]. The right inclusion is strict: $M_0 = M_{\pm 1} = a > 0$ gives $T(M) = a \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right) \succeq 0$, the moment matrix of the point mass $a\delta_0$, whereas a bounded nonnegative function with these coefficients would satisfy $\int_0^L m(x)(1 - \cos(2\pi x/L))\, dx = 0$, forcing $m = 0$ a.e. and contradicting $M_0 = a > 0$; singular $T(M)$ corresponds to atomic measures. For $T(M) \succ 0$ the Levinson–Durbin recursion gives the maximum-entropy autoregressive extension $f(\theta) = \varsigma^2 / |a(e^{i\theta})|^2$ (a rational density; ς^2 is the innovation variance) with a zero-free on the closed disk, matching $M_0, \dots, M_{K_{\max}}$; f is continuous, bounded and bounded away from zero, and scaling it to $\max f = 1$ does not change ρ: this proves the left inclusion, which is strict because $\mathcal{C}_+$ is not open: $M = 0$ (the zero mask) lies in $\mathcal{C}_+$ but not in $\{T \succ 0\}$. The positive-definite cone is dense in the semidefinite cone, so $\overline{\mathcal{C}_+} = \{T \succeq 0\}$. Equality of the optimal values: for any M with $T(M) \succeq 0$ and $\rho(M) < \infty$ (positive bright intensities), $M_\varepsilon = M + \varepsilon e_0$ satisfies $T(M_\varepsilon) = T(M) + \varepsilon I \succ 0$, so $M_\varepsilon \in \mathcal{C}_+$, and $\rho(M_\varepsilon) \to \rho(M)$ by continuity of ρ at points with positive bright intensities. □

Remark. In two dimensions block-Toeplitz positivity is necessary but not sufficient for the truncated trigonometric moment problem [24]; it still yields a valid relaxation.

Proposition 6 (Cutoff)**.** *If for every source point at most one collected order passes the pupil, then $Q_{x,z}$ is independent of x, $t_R = 1$, and uniform (μ, λ) is a certificate. For $d = 1$ the order k passes for the source point $s_j = (s_{jx}, s_{jy})$ iff $|k/L + s_{jx}/\lambda| \le \sqrt{\mathrm{NA}^2 - s_{jy}^2}/\lambda$, so the hypothesis holds iff no two integers lie in any of the intervals $[-s_{jx} - \sqrt{\mathrm{NA}^2 - s_{jy}^2},\ -s_{jx} + \sqrt{\mathrm{NA}^2 - s_{jy}^2}]\, L/\lambda$; a sufficient condition, under which only the zero order passes, is $L \max_j \left(|s_{jx}| + \sqrt{\mathrm{NA}^2 - s_{jy}^2}\right) < \lambda$, which for a disk illuminator of partial coherence σ is the classical cutoff $L < \lambda/((1+\sigma)\mathrm{NA})$. For $d = 2$ a sufficient condition is $\max(L_1, L_2)(\mathrm{NA} + \max_j |s_j|) < \lambda$. The radial condition $L(\mathrm{NA} + \max_j |s_j|) < \lambda$ is sufficient but not necessary in general (e.g. $\mathrm{NA} = \lambda = 1$, $L = 2/3$, $s = (0, 0.7)$: only the zero order passes although $L(\mathrm{NA} + |s|) = 1.13$).*

Proof. With a single order per source point the image is $\sum_j w_j |M_{k_j}|^2$, independent of x; uniform weights give $S = 0$ and $\sum_b \lambda_b = 1$. □

Remark. The converse fails for some targets: just above the cutoff only the orders $0, \pm 1$ interfere and the image is a sinusoid at the cell period, so a target whose bright and dark samples cannot be separated by such a sinusoid (e.g. $B = \{0, L/2\}$, $D = \{L/4, 3L/4\}$) still has $t_R = 1$. For dense lines $t_R < 1$ immediately above the cutoff, and for two-beam imaging of dense lines at pitch p the bound equals the closed form $(1 - \sin(2\pi\tau/p))/(1 + \sin(2\pi\tau/p))$ (verified to 2×10^{-5} in the test suite).

Lemma 7 (Complex versus real masks)**.** *Let $w(-s) = w(s)$ and $P_0(-g) = \overline{P_0(g)}$ (even pupil amplitude, odd aberration phase). Then $\tilde{M} = \mathrm{conj}(\Pi M)$ produces the same image as M at best focus, and with defocus $I_{\tilde{M},z} = I_{M,-z}$. Consequently, for a focus set with $Z = -Z$ and no even aberrations, $t_R(\mathcal{M}_{\mathbb{C}}) = t_R(\mathcal{M}_{\mathbb{R}})$ (take $X = \frac{1}{2}(MM^{\mathsf{H}} + \tilde{M}\tilde{M}^{\mathsf{H}})$). Even aberrations (astigmatism, spherical, Z_4) break the identity.*

Proof. Substituting $k \to -k$ in Eq. (2) and using $\overline{P_z(-g)} = P_{-z}(g)$ and $w(-s) = w(s)$ gives $I_{\tilde{M},z}(x) = I_{M,-z}(x)$; X is real-structured since $\mathrm{conj}(\Pi M M^{\mathsf{H}} \Pi) = \tilde{M}\tilde{M}^{\mathsf{H}}$, and its constraint values are averages over $z \in Z = -Z$. □

Lemma 8 (Vector imaging and aberrations)**.** *With the image-side plane wave of direction cosines (α, β, γ) carrying the mask-side polarization p rotated by the minimal rotation taking $\hat{z}$ to the propagation direction (for $p = \hat{x}$: $E = (1 - \alpha^2/(1+\gamma),\ -\alpha\beta/(1+\gamma),\ -\alpha)$, Rodrigues' formula; cf. [33]), the image is a sum of rank-one Hermitian forms over channels (source point, polarization, field component), so Theorems 1–3 apply verbatim; Lemma 7 extends to reflection-symmetric polarization ensembles since $E_{x,y}(-g) = E_{x,y}(g)$, $E_z(-g) = -E_z(g)$.*

Lemma 9 (Pixelation)**.** *For a mask piecewise constant on $N \ge F$ pixels per period (one dimension; pixel n covers $[nL/N, (n+1)L/N)$), $M_k = \mathrm{sinc}(k/N)\, e^{-i\pi k/N}\, \mathrm{DFT}_k(m)$ maps onto $\mathbb{C}^F$, so the pixel-domain and Fourier-domain relaxations coincide for the scale-free ratio problem; $N < F$ can only raise the bound.*

Remark (Sampled versus continuous constraints). The continuous problem has more constraints than the sampled one, so a certificate for the sampled problem is a valid lower bound for the continuous problem; tightness is a separate question. Numerically the interior samples carry zero dual weight and the bound is invariant to the interior spacing (§6.9).

Theorem 10 (Single-mask lower bound by branch and bound)**.** *Let $Q_1, \dots, Q_m \succeq 0$ (dark) and $P_1, \dots, P_n \succeq 0$ (bright) be real symmetric and $\rho(r) = \max_i q_i(r) / \min_j p_j(r)$. (i) $\rho^\star(\mathcal{M}_{\mathbb{R}}) = \inf$ over the faces $\Phi_i = \{r : r_i = 1, |r_j| \le 1\}$. (ii) For a box $\{c + \delta : |\delta_j| \le h_j\}$ and*

$q(r) = r^{\mathsf{T}}Qr$ *with* $Q \succeq 0$*:* $q(c) - 2|Qc| \cdot h \le q(r) \le q(c) + 2|Qc| \cdot h + h^{\mathsf{T}}|Q|h$*. (iii) Hence* $\inf_{box} \rho \ge \mathrm{LB} := \max_i \max(q_i(c) - 2|Q_i c| \cdot h, 0) / \min_j (p_j(c) + 2|P_j c| \cdot h + h^{\mathsf{T}}|P_j|h)$*, and a branch-and-bound over boxes covering* $\bigcup_i \Phi_i$ *returns, at any stopping time, a valid lower bound* $L = \min_{leaves} \mathrm{LB} \le \rho^\star(\mathcal{M}_{\mathbb{R}})$ *(leaves include the still-open boxes); on convergence* $L \ge (1 - \mathrm{tol})\, U$ *with* U *the incumbent.*

Proof. (i) Scale and sign invariance of ρ: every $r \neq 0$ has a coordinate of maximal modulus, which can be scaled to $+1$. (ii) $q(c+\delta) = q(c) + 2(Qc)\cdot\delta + \delta^{\mathsf{T}}Q\delta$ with $0 \le \delta^{\mathsf{T}}Q\delta \le h^{\mathsf{T}}|Q|h$. (iii) Interval arithmetic and the covering argument. □

5 Algorithms

Bound. The source quadrature is first compressed exactly into at most F coherent channels by the eigendecomposition of the $F \times F$ transmission-cross-coefficient matrix $T_z = \sum_j w_j P_z(f + s_j/\lambda) P_z(f + s_j/\lambda)^{\mathsf{H}}$ (the sum-of-coherent-systems decomposition [30]; images agree to 10^{-15}), then $Q_{x,z}$ is built on the edge-offset sample set and the real SDP is solved (CLARABEL [34]; SCS [35] as fallback).

Rigorization and exact verification. Extract the duals (μ, λ) and find by bisection in double precision the largest $\kappa \le 1$ with $\lambda_{\min}(S_D - \kappa S_B) \ge 10^{-10}\|S_D - \kappa S_B\|$, $S_D = \sum_d \mu_d Q_d$, $S_B = \sum_b \lambda_b Q_b$ (a heuristic safeguard: the margin is relative to a difference that can suffer cancellation, and the forms Q carry rounding errors of order 10^{-15}). The reported bound is $\kappa \sum_b \lambda_b$. The certificate is then verified independently of this computation: the matrix $\tilde{S} = \mathrm{Re}\, U^{\mathsf{H}}(S_D - \kappa S_B)U$ is *rebuilt* in 50-digit arithmetic from the model equations (1)–(2) and the instance data (cell, wavelength, NA, index, focus set, source points and weights, sample coordinates; all exact as the double-precision numbers that define the instance), using the same (μ, λ, κ), and $\tilde{S} - 10^{-30} I$ is Cholesky-factorized at 50 digits. This proves $\lambda_{\min}(\tilde{S}) \ge 10^{-30}$ up to the 50-digit rounding error ($\lesssim F^2\, 10^{-48}\|\tilde{S}\|$), hence Theorem 1 applies to the stated discrete instance. The minimum eigenvalue of the rebuilt matrix is reported with each certificate (§6.9). What is not proved is a bound for a *different* discretization (finer source quadrature, other sample points), and no interval arithmetic is used.

Primal. Gaussian rounding from X; analytic-gradient L-BFGS on a log-sum-exp surrogate of $\log \rho$ (common dose across foci); vectorized SLSQP minimax polish; multistart; optimal-face exploration (random PSD objectives over the optimal face) for low-rank optima; explicit two-exposure refinement.

Single-mask certification. (a) Theorem 10 with batched boxes; error terms are multiplied by $1 + 10^{-9}$ and 10^{-12} is subtracted (a heuristic safeguard, not a proof of rounding control). (b) An SDP-bounded variant: to certify $\rho^\star(\mathcal{M}_{\mathbb{R}}) \ge T$ the set $\{r \in \mathrm{box} :\ r^{\mathsf{T}}(Q_d - T P_b) r \le 0\ \forall (d, b)\}$ is shown empty through the Shor relaxation $[1, r^{\mathsf{T}}; r, X] \succeq 0$ with reformulation–linearization box cuts, solved as "minimize the common slack"; each node is pruned only if an explicit dual certificate $(y \ge 0,\ \alpha,\ \beta)$ with $\sum_m y_m G_m + \alpha E_{00} + \beta E_{ii} \succeq 0$ and $\sum_m y_m c_m - \alpha - \beta > 0$ is verified after the solve (weak duality), so solver inaccuracy cannot produce a false pruning; restricting the pairs (d, b) to the active supports of the global certificate keeps the relaxation valid. (c) Lasserre-2 moment relaxations [22] are reported as corroboration only (solver tolerance limited).

Variants. SMO relaxation (one PSD block per source point: unlimited exposures with coherent point sources); doubly-nonnegative (DNN) relaxation for nonnegative masks on a pixel grid; complex-mask relaxation; multi-focus; vector and aberrated pupils; pixel-integrated masks.

Complexity. $F \approx (2(1+\sigma)\mathrm{NA}L/\lambda)^d$; the SDP has an $F \times F$ matrix variable and $|B| + |D|$ constraints per focus. Solve times are below 2.4 s for $F \le 15$ and 1324–2243 s for $F = 61$ with 1,273 constraints.

6 Experiments

Settings. DUV: $\lambda = 193$ nm, NA = 1.35, defocus in water ($n = 1.44$); EUV: $\lambda = 13.5$ nm, NA = 0.33, vacuum. Illuminators are area-weighted midpoint-grid quadratures of the illuminator aperture on a 401×401 grid over its bounding square: coherent (one on-axis point); conventional $\sigma \in \{0.5, 0.7, 0.9\}$ (disk; 126,309 points for $\sigma = 0.7$); dipole $\sigma_c \in \{0.7, 0.8\}$ (two circular poles of radius 0.15 NA on the x axis, 121×121 grid per pole); annular 0.6/0.9 (DUV) and 0.5/0.9 (EUV); quasar (four 30° sectors of the annulus). The pupil cutoff makes the integrand discontinuous inside the source, so the bound converges only like $1/n$ with the grid size n; at $n = 401$ it is converged to about 10^{-4} (§6.9), and the coherent-mode compression of §5 makes the point count immaterial for cost. The per-source SMO relaxation uses a coarse 21×21 rule (349 points), since it carries one matrix variable per source point. Edge tolerance τ is 10% of the minimum CD (5 nm for the 50 nm one-dimensional features). "Bound" is always the rigorized certificate value and "best single mask" the best feasible real mask found (§3.3); EL is the full-range value (4). All tables are generated from the clean reproduction.

6.1 Resolution cutoff and two-beam limit

For dense lines the bound is exactly 1 for pitches below $\lambda/((1+\sigma)\mathrm{NA}) = 84.1$ nm at $\sigma = 0.7$ and drops below 1 immediately above (Fig. 1, right). With an on-axis dipole at the optimal angle the bound for dense lines equals the two-beam closed form of Proposition 6's remark (0.52786 at pitch 90 nm, $\tau = 4.5$ nm). In EUV with ±50 nm focus (Table 2, Fig. 1, left) the 20 nm pitch ($k_1 = 0.244$) is certified unprintable with any mask under both illuminators; at 22 nm pitch the $\sigma = 0.9$ disk reaches a full-range EL of only 0.055 (only a sliver of the disk produces two-beam interference; the best single mask attains 0.055) and the finite-pole dipole bound is 0.7404. Of the 14 EUV cases, 13 have ratio gaps below 10^{-4} and the largest gap is 3.63% (40 nm pitch, finite-pole dipole). Because EL $\approx 2(1-\rho)$ near $\rho = 1$, a ratio gap of a few percent can bracket the exposure latitude loosely (there the achieved EL is 0.508 against a bound of 0.541), so Table 2 reports both the EL at the bound and the EL achieved. These EUV values are aerial-image entitlements of the thin-mask scalar model and ignore mask shadowing and three-dimensional mask effects, flare, resist blur and stochastics; they are necessary, not sufficient, conditions for printability.

6.2 One-dimensional feature library

Dense-line cases are tight to 0.19% or better and dipole cases to 1.27%; the conventional-illumination gaps are 0.0% (mixed) and 0.6% (iso-dense); coherent illumination shows large gaps (iso-dense 19.1%, mixed 9.6%), and the three-focus cases show gaps of 1%–9% (Table 3). Certified full-range exposure latitudes at $\tau = 5$ nm range from 0.27 to 0.62 at best focus and from 0.18 to 0.36 over ±80 nm. Figure 2 shows a certificate for the iso-dense target: the dual weights concentrate on the edge-offset samples.

6.3 Random one-dimensional targets

On 109 random targets (1–4 segments of width ≥ 40 nm in 300–600 nm cells; $\tau = 5$ nm), 1 instance(s) are near-degenerate (certified bound below 10^{-3}, i.e. a mask makes every dark sample almost exactly dark; best mask found $\rho \le 0.018$), for which a relative gap is meaningless. On the remaining 108 the best single mask matches the certified bound to 10^{-4} in 74.1% of

Table 2: EUV dense lines ($\lambda = 13.5\,\text{nm}$, NA $= 0.33$, vacuum, focus set $\{-50, 0, +50\}\,\text{nm}$, $\tau = 10\%$ CD $= 0.05$ pitch; the periodic cell contains four pitches, $L = 4\times$pitch, with four lines of width pitch/2). Columns: pitch; k_1; illuminator; collected orders F; bound ρ; full-range EL at the bound (upper bound on the achievable EL); best single mask ρ and its EL (achieved); ratio gap.

Pitch (nm)	k_1	Source	F	Bound ρ	Max EL	Best single ρ	EL (best)	Gap
20	0.244	conv0.9	7	1.0000	0.000	1.0000	0.000	0.00%
20	0.244	dipole0.8	7	1.0000	0.000	1.0000	0.000	0.00%
22	0.269	conv0.9	9	0.9467	0.055	0.9467	0.055	0.00%
22	0.269	dipole0.8	9	0.7404	0.298	0.7404	0.298	0.00%
24	0.293	conv0.9	9	0.8405	0.173	0.8405	0.173	0.00%
24	0.293	dipole0.8	9	0.5714	0.546	0.5714	0.546	0.00%
26	0.318	conv0.9	9	0.7806	0.246	0.7806	0.246	0.00%
26	0.318	dipole0.8	9	0.5457	0.588	0.5457	0.588	0.00%
28	0.342	conv0.9	11	0.7407	0.298	0.7407	0.298	0.00%
28	0.342	dipole0.8	11	0.5551	0.572	0.5551	0.572	0.00%
32	0.391	conv0.9	11	0.6891	0.368	0.6891	0.368	0.00%
32	0.391	dipole0.8	13	0.5902	0.515	0.5902	0.515	0.00%
40	0.489	conv0.9	15	0.6284	0.456	0.6284	0.456	0.00%
40	0.489	dipole0.8	15	0.5740	0.541	0.5948	0.508	3.6%

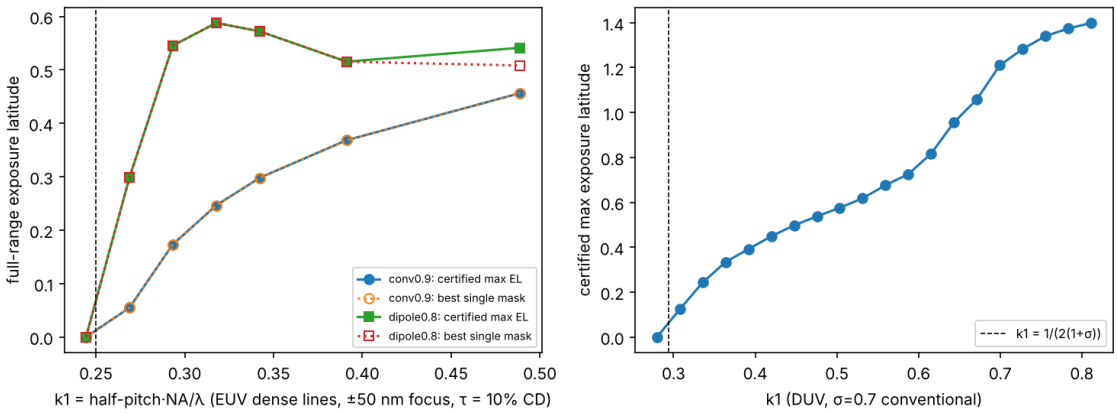


Figure 1: Certified maximum full-range exposure latitude versus k_1 for dense lines. Left: EUV with $\pm 50\,\text{nm}$ focus under the $\sigma = 0.9$ disk and the $\sigma_c = 0.8$ finite-pole dipole; certified bound (upper bound on EL) and best single mask found (achieved EL) agree within a ratio gap of 3.63% at every point (Table 2 gives both); the dashed line marks $k_1 = 0.25$. Right: DUV, conventional $\sigma = 0.7$ disk; the dashed line is the classical cutoff $k_1 = 1/(2(1+\sigma))$, below which the certified exposure latitude is exactly zero (Proposition 6).

instances, to 1% in 85.2% and to 5% in 93.5% (median gap 4.8×10^{-7}, maximum 55.7%; Fig. 3). Exact tightness is source-dependent (100% for $\sigma = 0.9$, 62% for the dipole, 57% for coherent and 68% for $\sigma = 0.5$ illumination); the 7 gaps above 5% have numerical rank up to 3, consistent with Theorem 3.

6.4 Certified multi-exposure gains

The interval branch-and-bound of Theorem 10 converges in 0.3–3.2 s for $F = 5$ and proves the gains of Table 4 (Fig. 4, left); each optimal face has rank 2, so two exposures suffice. For the mixed-width coherent target ($F = 7$) a 30-million-node run (58 min) leaves a 0.63% gap between $L = 0.6399$ and $U = 0.6439$ and certifies a gain of 8.9%. The SDP-bounded variant was run with thresholds T chosen strictly between the ratio of the feasible multi-exposure ensemble and the single-mask incumbent (the only informative range: a threshold below $\bar{t}$ certifies nothing,

Table 3: One-dimensional feature library (DUV, $\lambda = 193\,\text{nm}$, NA $= 1.35$, water-immersion defocus; $\tau = 5\,\text{nm}$, except 4.5 nm for the 90 nm-pitch dense lines). F: collected orders; foci: $|Z|$, with $\pm 80\,\text{nm}$ denoting $\{-80, 0, +80\}\,\text{nm}$; bound ρ: rigorized certificate; max EL: full-range exposure latitude (4) at the bound; best single ρ: best feasible real mask found, with its full-range EL (achieved); gap = (best single − bound)/bound. "iso-dense": three 50 nm lines at 100 nm pitch plus an isolated line in a 600 nm cell; "mixed": 40, 80 and 160 nm lines in a 500 nm cell.

Case	F	Foci	Bound ρ	Max EL	Best single ρ	EL (best)	Gap
iso-dense, coherent	9	1	0.5359	0.604	0.6383	0.442	19.1%
mixed, coherent	7	1	0.5875	0.520	0.6439	0.433	9.6%
iso-dense, $\sigma = 0.7$	15	1	0.7163	0.331	0.7208	0.325	0.63%
mixed, $\sigma = 0.7$	11	1	0.6812	0.379	0.6812	0.379	0.00%
iso-dense, dipole 0.8	17	1	0.5658	0.555	0.5730	0.543	1.3%
mixed, dipole 0.8	13	1	0.6020	0.497	0.6020	0.497	0.00%
iso-dense, $\sigma = 0.7$, $\pm 80\,\text{nm}$	15	3	0.8305	0.185	0.8610	0.149	3.7%
mixed, $\sigma = 0.7$, $\pm 80\,\text{nm}$	11	3	0.8370	0.177	0.8451	0.168	0.97%
iso-dense, dipole 0.8, $\pm 80\,\text{nm}$	17	3	0.6940	0.361	0.7566	0.277	9.0%
dense 90, $\sigma = 0.7$	9	1	0.7632	0.269	0.7646	0.267	0.19%
dense 90, dipole 0.8	9	1	0.5279	0.618	0.5279	0.618	0.00%
dense 90, coherent	5	1	0.5279	0.618	0.5279	0.618	0.00%

Table 4: Separation of single- and multi-exposure optima (DUV). SDP bound: t_R (any number of exposures); B&B L: certified lower bound on the single-mask optimum $\rho^\star(\mathcal{M}_\mathbb{R})$ (Theorem 10); best single U: incumbent, $L \le \rho^\star(\mathcal{M}_\mathbb{R}) \le U$; nodes explored; converged: $L \ge (1 - 10^{-3})U$; certified gain $= (L - \bar{t})/\bar{t}$, where $\bar{t} \ge t_R$ is the ratio realized by the multi-exposure ensemble of the SDP solution ("none" when $L \le \bar{t}$, i.e. the run is uninformative). "random n" are instances of the random one-dimensional set.

Case	F	SDP bound	B&B L	Best single U	Nodes	Converged	Certified gain
random 82 (coherent)	5	0.3597	0.4061	0.4066	40,383	yes	12.9%
random 91 (coherent)	5	0.6049	0.6775	0.6782	76,027	yes	12.0%
random 110 (coherent)	5	0.3464	0.4033	0.4037	17,413	yes	16.4%
random 50 (coherent)	5	0.3141	0.3343	0.3347	28,517	yes	6.4%
mixed (coherent)	7	0.5875	0.6399	0.6439	30,029,005	no (0.63% gap)	8.9%
iso-dense (coherent)	9	0.5359	0.1496	0.6383	6,005,751	no (76.56% gap)	none
mixed, $\sigma = 0.7$, $\pm 80\,\text{nm}$	11	0.8370	0.0000	0.8451	6,007,797	no (100.00% gap)	none

one above U cannot be certified). It certifies: none; it did not terminate within 6,000 nodes for: iso-dense (coherent): $T = 0.56$; mixed, $\sigma = 0.7$, $\pm 80\,\text{nm}$: $T = 0.84$; mixed (coherent): $T = 0.6$. Lasserre-2 single-mask bounds agree with the best single masks to 7×10^{-5} (solver tolerance; CLARABEL 0.6439516 versus SCS 0.6439091 on the mixed target) and corroborate, without certifying, gains of 19.1% ($F = 9$, iso-dense coherent) and 1.0% ($F = 11$, three-focus mixed target).

6.5 Nonnegative versus phase-shift masks and pattern-error ILT

Table 5 compares the certified optima of real (phase-shift) and nonnegative masks with a standard pattern-error ILT on the iso-dense, mixed and dense-120 targets ($\sigma = 0.7$, $\tau = 5\,\text{nm}$). Comparing two lower bounds certifies nothing, so the penalty of nonnegative masks is bracketed: with L_+ the DNN lower bound on nonnegative masks on a 5 nm pixel grid, U_R the best feasible PSM ratio, L_R the certified PSM bound and U_+ the best feasible nonnegative pixel mask found (bounded L-BFGS on the ratio surrogate from eight starts), $L_+/U_R - 1 \le \rho_+^\star/\rho_R^\star - 1 \le U_+/L_R - 1$. The certified lower bounds are 2.6%–11.4% and the upper bounds 3.6%–17.6% (the Toeplitz cone of Theorem 5 would extend the lower bound to all nonnegative masks). Pixel- and Fourier-domain PSM relaxations agree to 1×10^{-3}. A pattern-error ILT (sigmoid resist, open-frame-normalized intensity, threshold 0.3, best of six settings, 1,500 Adam steps; gradient checked

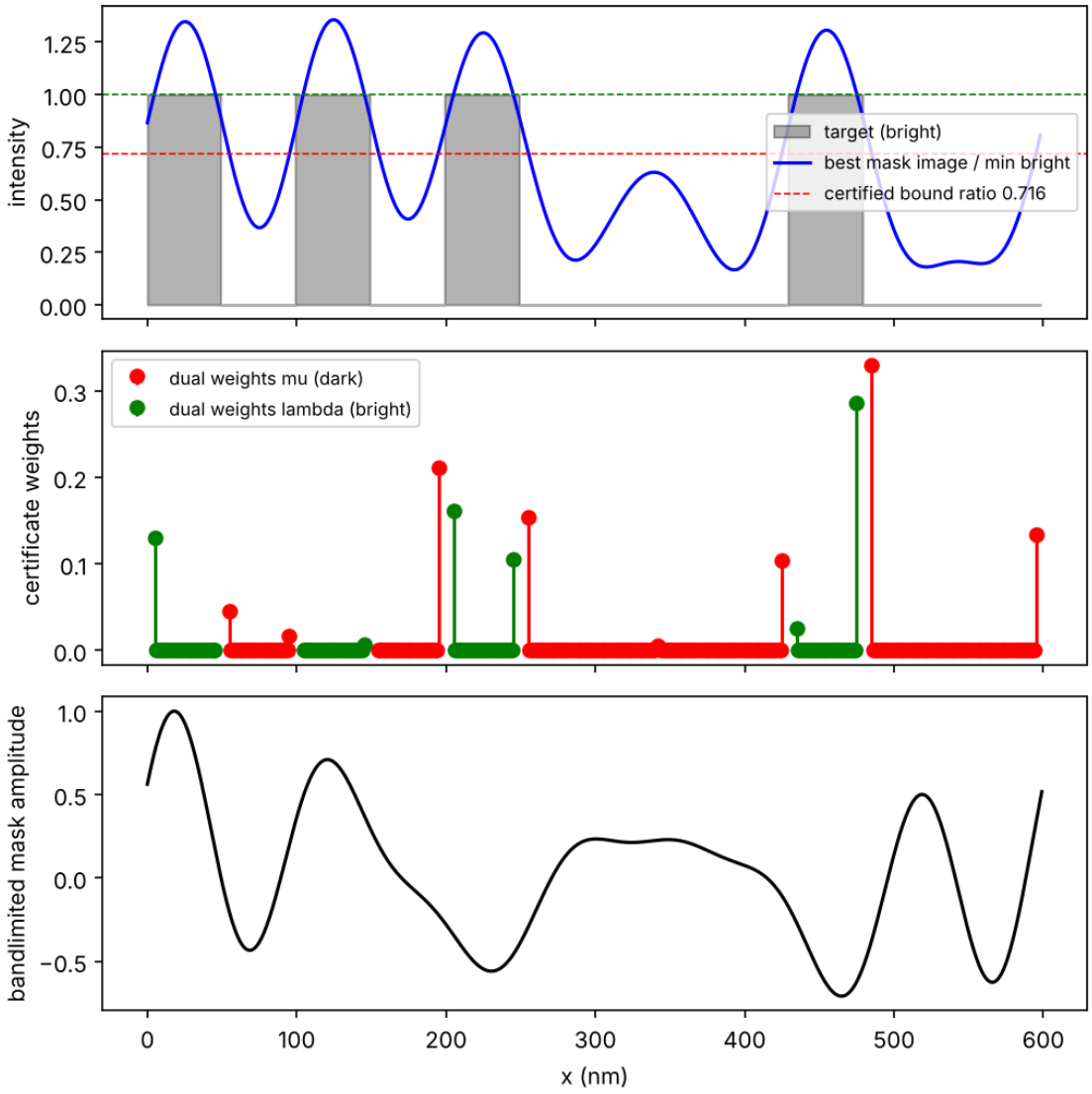


Figure 2: Certificate for the iso-dense target under the conventional $\sigma = 0.7$ disk (DUV; Table 3). Top: target (gray) and the image of the best single mask normalized to its minimum bright intensity; the dashed red line is the certified bound. Middle: dual weights μ_d (dark, red) and λ_b (bright, green), concentrated on the edge-offset samples. Bottom: band-limited amplitude of the best mask (negative values are π-phase regions).

against finite differences in the test suite) reaches a ratio whose suboptimality relative to the PSM optimum lies between $\rho_{\mathrm{ILT}}/U_R - 1$ and $\rho_{\mathrm{ILT}}/L_R - 1$, i.e. 19%–32% (the brackets are tight because $L_R \approx U_R$); refining the ILT mask on the ratio objective removes part of this gap.

6.6 Source–mask entitlement

For the iso-dense target the per-source-point relaxation (unlimited exposures, each with a coherent point source of the coarse 349-point rule) equals the coherent-illumination relaxation (0.5359) both for the $\sigma = 0.7$ disk (fixed-source bound 0.7183) and for the $\sigma = 0.9$ disk (0.7202); for the dipole the two coincide (0.5760). Thus no illumination supported on these points improves the *multi-exposure* entitlement beyond on-axis coherent light. For a single exposure the statement is different: the best single coherent mask (Table 3) is worse than the best single dipole mask, because the coherent relaxation is not tight.

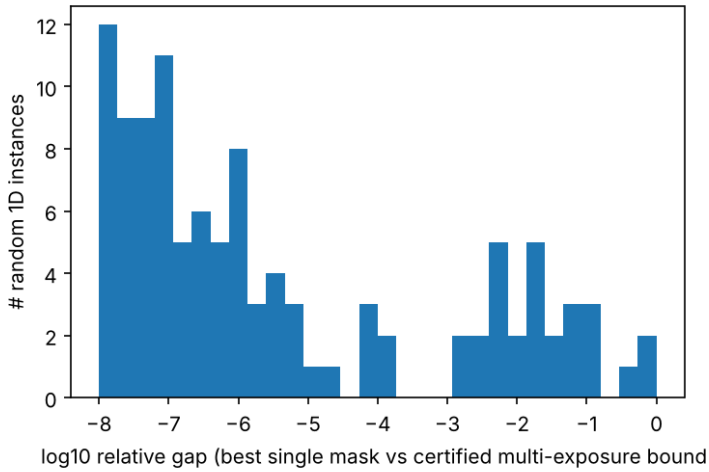


Figure 3: Histogram of $\log_{10}$ of the relative gap between the best single mask and the certified multi-exposure bound over the 109 random one-dimensional targets. The population is bimodal: most instances are tight to solver precision, while the instances with gaps of 10^{-2}–10^{-1} have higher-rank optima (Theorem 3).

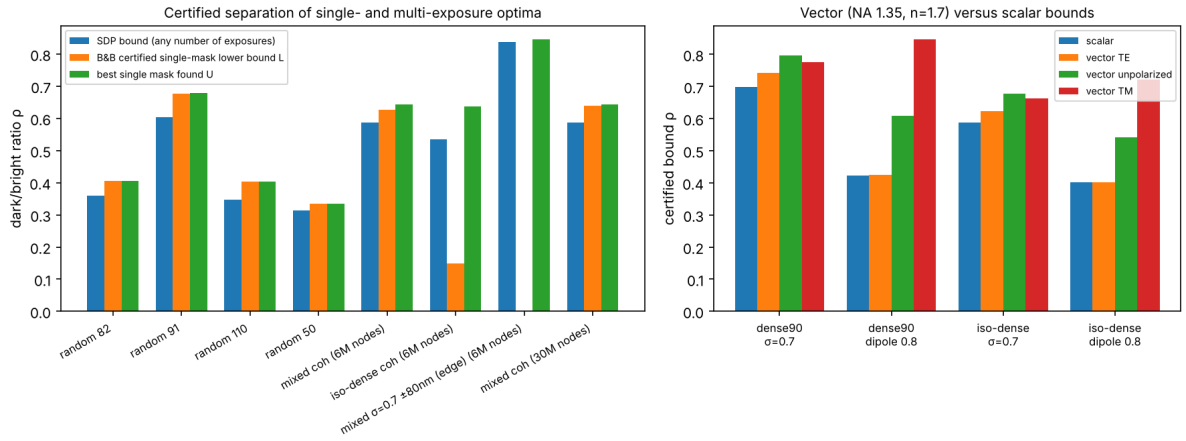


Figure 4: Left: SDP bound t_R, certified single-mask lower bound L and best single mask U for the cases of Table 4 (the last three columns are the non-converged runs, including the 30-million-node run on the mixed target); a gap between the first and second bar proves that multiple exposure strictly improves the ratio. Right: bounds under the scalar model and under the vector model (NA = 1.35, resist index 1.7) for TE, unpolarized and TM illumination on four one-dimensional targets (§6.9).

Table 5: Certified real-mask (PSM) and nonnegative-mask optima versus a pattern-error ILT (DUV, conventional $\sigma = 0.7$ disk, $\tau = 5\,\mathrm{nm}$). All entries are ratios ρ (lower is better). L_R: certified PSM bound; U_R: best feasible PSM mask; L_+: DNN lower bound for nonnegative masks on a 5 nm pixel grid; U_+: best feasible nonnegative pixel mask; penalty bracket: $[L_+/U_R - 1,\ U_+/L_R - 1]$, a certified interval for $\rho_+^\star/\rho_R^\star - 1$; ILT (PSM): ratio reached by the pattern-error ILT mask and its suboptimality bracket $[\rho_{\mathrm{ILT}}/U_R - 1,\ \rho_{\mathrm{ILT}}/L_R - 1]$.

Target	L_R	U_R	L_+	U_+	Penalty bracket	ILT (PSM)	ILT suboptimality
iso-dense	0.7163	0.7208	0.7529	0.7976	[4.5%, 11.4%]	0.8613	[19.5%, 20.3%]
mixed	0.6812	0.6812	0.7587	0.8008	[11.4%, 17.6%]	0.8975	[31.8%, 31.8%]
dense 120	0.6735	0.6767	0.6942	0.6978	[2.6%, 3.6%]	0.8261	[22.1%, 22.7%]

Table 6: Two-dimensional layouts (DUV). Contact arrays: pitch 160, 120 and 100 nm with square holes of width 70, 50 and 44 nm (0.44 pitch), $\tau = 10\%$ of the hole width (7, 5 and 4.4 nm); T-junction and line-end cells: 320 nm cell, 60 nm lines, $\tau = 6$ nm. Columns: F; bound ρ; full-range EL at the bound; best single mask ("strong": 12-start search with SLSQP polish, and 60-iteration basin hopping for the quasar T-junction) and its EL; gap; wall-clock time of the relaxation solve in seconds.

Layout	F	Bound ρ	Max EL	Best single ρ	EL (best)	Gap	SDP time (s)
contacts 160, annular	13	0.8040	0.217	0.8221	0.195	2.25%	2.4
contacts 160, $\sigma = 0.5$	9	0.8383	0.176	0.8384	0.176	0.01%	1.8
contacts 120, annular	9	0.8749	0.133	0.8765	0.132	0.18%	0.9
contacts 120, quasar	9	0.8385	0.176	0.8394	0.175	0.11%	1.5
contacts 100, quasar	5	0.9782	0.022	0.9782	0.022	0.00%	0.8
T-junction 320, annular (strong)	61	0.8050	0.216	0.8066	0.214	0.20%	2242.7
T-junction 320, quasar (strong)	49	0.7817	0.245	0.7911	0.233	1.20%	794.7
line-ends 320, annular (strong)	61	0.8677	0.142	0.8681	0.141	0.04%	1323.8

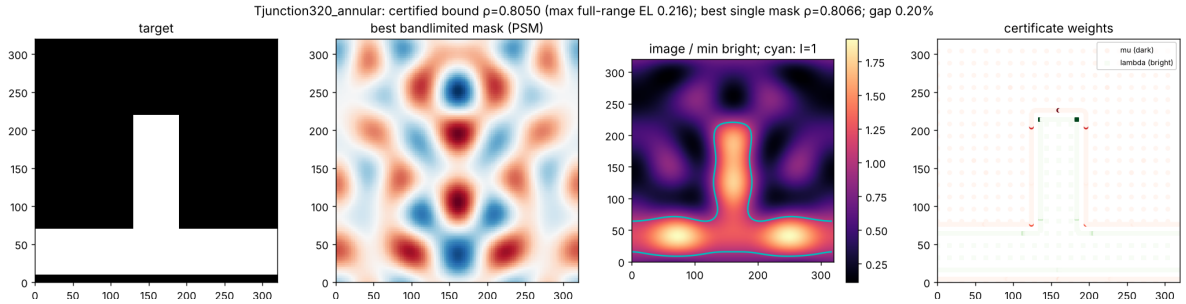


Figure 5: T-junction in a 320 nm cell under annular illumination (DUV, $\tau = 6$ nm; axes in nm): target (white = bright); best band-limited real mask of the strong search (signed amplitude); normalized image with the $I = 1$ contour (cyan); certificate weights μ_d (dark) and λ_b (bright), which localize the limiting corners and the line-end. The panel title states the certified bound and the gap of the mask shown.

6.7 Two-dimensional layouts

Contact arrays are certified tight to 2.25% or better, with numerical rank at most 1 (Table 6). For the 320 nm cells ($F = 61$, 1,273 constraints) the strong search closes the annular T-junction and the line-end gaps to 0.20% and 0.04% (Fig. 5); the quasar T-junction retains a 1.2% gap that we could neither close nor certify as a multi-exposure gain. On 24 random layouts of one to three rectangles in 200–240 nm cells (annular, quasar and $\sigma = 0.5$ illumination; $F = 13$–37) the single-mask search matches the bound to 10^{-3} in 45.8% and to 5% in 79.2% of cases; the remaining gaps (up to 11%) are unresolved upper bounds on the multi-exposure benefit (Fig. 6, left). EUV contact arrays (32–44 nm pitch, ±40 nm focus, $\tau = 10\%$ CD) are tight to 0.00% with certified full-range EL from 0.09 to 0.29 (Fig. 6, right), subject to the EUV caveats above.

Corner convention. The two-dimensional sample sets enforce the edge tolerance along the whole offset contour, including the circular arcs around convex corners and the inner corners of contact holes, where the aerial image necessarily rounds. The resulting entitlements are therefore conservative relative to edge-placement checks that exclude a zone around corners, as is common practice. Table 7 quantifies this: dropping the offset samples whose foot point lies within 0.15 or 0.25 minimum CD of a corner raises the certified full-range EL by a factor 1.6–4.6 (0.25 CD) on the 5 layouts tested (e.g. from 0.22 to 0.37 for the 160 nm contacts); all other two-dimensional numbers in this paper use the full contour.

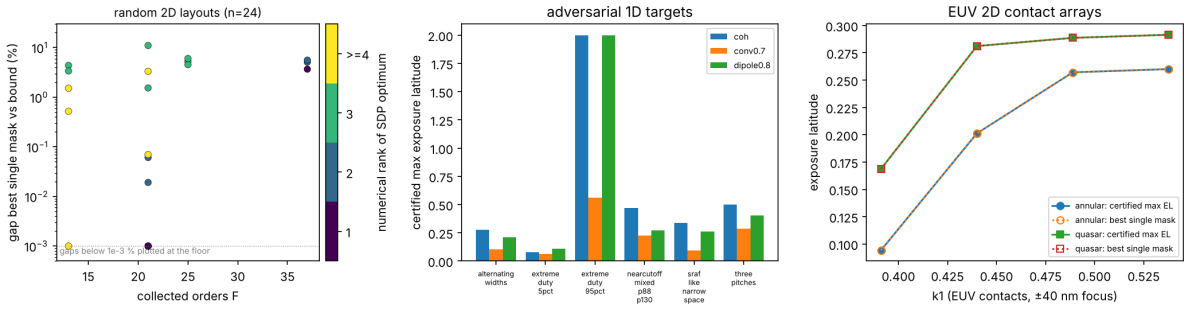


Figure 6: Expanded dataset. Left: gap between the best single mask and the certified bound (percent, logarithmic) versus F for the random two-dimensional layouts; color encodes the numerical rank of the SDP optimum. Middle: certified maximum full-range EL of the adversarial one-dimensional targets (§6.8) under coherent, $\sigma = 0.7$ disk and finite-pole dipole illumination. Right: EUV two-dimensional contact arrays under annular and quasar illumination; certified EL and best single mask coincide.

Table 7: Sensitivity of the two-dimensional certified bounds to a corner-exclusion zone (DUV; τ as in Table 6). Offset samples whose foot point lies within c of a boundary corner are dropped; $c = 0$ is the convention of all other tables. Columns: bound ρ and full-range EL at the bound for $c = 0$, $c = 0.15$ CD and $c = 0.25$ CD (a larger zone leaves no edge sample on the 70 nm contacts).

Layout	τ (nm)	ρ ($c = 0$)	EL	ρ (0.15 CD)	EL	ρ (0.25 CD)	EL
contacts 160, annular	7	0.8040	0.217	0.7666	0.264	0.6895	0.368
contacts 120, quasar	5	0.8385	0.176	0.8065	0.214	0.7370	0.303
contacts 100, quasar	4.4	0.9782	0.022	0.9490	0.052	0.9035	0.101
T-junction 320, annular	6	0.8050	0.216	0.7591	0.274	0.7046	0.347
line-end 320, annular	6	0.8677	0.142	0.8430	0.170	0.7777	0.250

6.8 Adversarial one-dimensional targets

Near-cutoff mixed pitches, SRAF-like 20 nm spaces, 5%/95% duty cycles, alternating widths and three-pitch mixtures (Fig. 6, middle; $\tau = 10\%$ CD) are tight under dipole illumination, mostly tight under the $\sigma = 0.7$ disk, and show gaps of 10%–39% in 4 of six coherent cases, the regime in which multiple exposure helps most. The isolated 20 nm space under coherent or dipole light has a certified bound of ≈ 0: a phase-edge mask makes the sampled dark points exactly dark. This is a statement about the sampled constraint set, not about the printed width of a phase-edge line, which is dose-determined.

6.9 Robustness, solver validation and physical sensitivity

The bound is invariant to the interior sample spacing (0.71618–0.71626 for spacings of 16 to 1 nm) and monotone in τ (full-range EL: $\tau = 2$ nm: 0.13, $\tau = 2.5$ nm: 0.17, $\tau = 4$ nm: 0.27, $\tau = 5$ nm: 0.33, $\tau = 8$ nm: 0.52, $\tau = 12$ nm: 0.75, $\tau = 16$ nm: 0.95, $\tau = 20$ nm: 1.10). Quadrature convergence of the $\sigma = 0.7$ disk (iso-dense target): 21^2 grid (349 pts): 0.71830, 41^2 grid (1,313 pts): 0.71443, 61^2 grid (2,933 pts): 0.71678, 101^2 grid (8,021 pts): 0.71638, 201^2 grid (31,757 pts): 0.71624, 401^2 grid (126,309 pts): 0.71625, 801^2 grid (503,973 pts): 0.71619 (the 401×401 rule used throughout differs from the finest, 801×801, by 0.008%). This residual is a discretization uncertainty that solver certification does not cover: every certificate is a statement about the stated discrete illuminator, sample set and focus set, and only an estimate, not a bound, is available for the continuous illuminator. Because the continuous printing problem has more constraints than the sampled one, a bound for the sampled problem remains a valid bound for the continuous geometry under the same illuminator, but the tightness statements (best masks) hold only for the sampled constraints: zero dual weight on interior

Table 8: Certified SDP bound (real masks) versus the generalized-eigenvalue (Rayleigh-quotient) bound and the complex-mask relaxation (DUV, $\tau = 5\,\mathrm{nm}$; dense 90: 4.5 nm). All entries are lower bounds on ρ (higher is tighter).

Case	Source	F	SDP (real)	Rayleigh	SDP (complex)
iso-dense	coherent	9	0.5359	0.1011	0.5359
iso-dense	$\sigma = 0.7$	15	0.7163	0.2526	0.7163
iso-dense	dipole 0.8	17	0.5658	0.1205	0.5658
mixed	coherent	7	0.5875	0.0044	0.5876
mixed	$\sigma = 0.7$	11	0.6812	0.0370	0.6812
mixed	dipole 0.8	13	0.6020	0.0141	0.6020
dense 90	coherent	5	0.5279	0.1925	0.5279
dense 90	$\sigma = 0.7$	9	0.7632	0.5253	0.7632
dense 90	dipole 0.8	9	0.5279	0.1925	0.5279

samples does not exclude violations between samples. The Rayleigh-quotient bound is 1.5–135× weaker (Table 8), and complex- and real-mask relaxations coincide to 2×10^{-5} on all 9 cases (Lemma 7). Every record of every experiment satisfies best single mask $\geq$ bound (0 violations), the consistency check that an invalid certificate would fail. All 143 bounds of the verification set were recomputed in double precision and their computed certificate matrices factorized by a 50-digit Cholesky (143 succeed, 0 degenerate with $S = 0$, 0 fail). Independently, for 143 of these instances the certificate matrix was rebuilt in 50-digit arithmetic from the model equations (§5): 143 are positive definite with margin 10^{-30} (minimum eigenvalue of the rebuilt matrix between 2×10^{-20} and 1×10^{-8}), 0 are zero certificates, 0 fail; for 1 instance(s) the double-precision scale κ was not safe for the exact matrix (a near-zero certificate at the EUV cutoff) and κ was lowered by an exact bisection at 50 digits, changing the bound by less than 10^{-15}. The reported bounds are the exactly verified values; on the 142 non-degenerate instances the rigorization scale is at least 0.992 and below 0.9999 in 4 of them; on the near-degenerate random instance the interior-point dual is unusable and the certificate is scaled down to 6×10^{-3} (a valid, nearly vacuous bound), which the local search confirms by reaching $\rho \leq 0.018$. CLARABEL and SCS agree to a median of 4.8×10^{-9}; on 4 near-degenerate instances (bounds up to 0.108) the first-order solver SCS is off by up to 0.03, which the certificate check exposes. Under the vector model ($n_{\mathrm{resist}} = 1.7$, Fig. 4, right) the one-dimensional TE bounds change by 0% to 6%, the unpolarized bounds by 14% to 44% and the TM bounds by 11% to 101% relative to the scalar model (the classical TM contrast loss at NA = 1.35), with gaps at most 1.7%; on the two-dimensional contacts the unpolarized bounds rise by 5%–7% and the TE bounds by 1%–1%. The vector model omits Fresnel transmission and thin-film effects, which rescale channel amplitudes without changing the structure of Lemma 8. Aberrations of 0.02–0.1λ move the iso-dense bound by up to 2%; pixelation at $N \geq F$ leaves it unchanged to 1×10^{-8} and coarser pixels raise it (50 nm: 0%; 67 nm: 8%), as Lemma 9 predicts.

7 Discussion

The certificates turn folklore into checkable statements. Local optimization of the exposure-latitude objective is certified near-optimal (within 10^{-4} to 10^{-2}) for most one-dimensional targets and for the two-dimensional contact arrays. When a gap remains, Theorem 3 identifies t_R with the multi-exposure optimum but not the local-search result with the single-mask optimum: the gap is an upper bound on the multi-exposure benefit that may also contain optimizer shortfall. A strict benefit is established only where a certified single-mask lower bound exceeds the ratio of a feasible ensemble, which the converged branch-and-bound runs do on concrete coherent-illumination targets; the unresolved gaps (quasar T-junction, large-F random layouts)

are not evidence either way. On-axis coherent light with real masks is the multi-exposure entitlement for the iso-dense target, and nonnegative masks forfeit a certified 2.6%–11.4% (at most 3.6%–17.6%). Pattern-error ILT is far from EL-optimal, which argues for EL or contrast objectives, or for the certificate as a stopping criterion. The dual weights localize the limiting geometry. The structural conclusions survive the vector model and moderate aberrations; the numerical entitlements depend on the illuminator quadrature, the defocus medium and the edge tolerance, all of which are stated with each table. Finally, the relaxation gap has a physical meaning and is informative rather than merely a weakness.

8 Limitations

The model is a thin-mask periodic model: no three-dimensional mask electromagnetics (shadowing and best-focus shifts in EUV), no resist chemistry or stochastics, no flare, no mask error, and no dose or MEEF constraint (the ratio is scale-free; a dose-aware variant with $\operatorname{Tr} X \le 1$ remains an SDP). Constraints are sampled with an explicit edge tolerance, and the EL objective fixes one threshold across the focus set; constraints linear in intensity (slope, dose-to-size) lift identically, but two-sided CD constraints and Boolean multi-patterning do not. The DNN relaxation is only an outer approximation in two dimensions and applies to pixelated nonnegative masks. "Certified" means: exact in exact arithmetic and verified in 50-digit arithmetic from the model equations for the stated discrete instance (no interval arithmetic, and no statement about other discretizations of the illuminator or of the geometry). The interval branch-and-bound converges only for $F = 5$ and is informative to $F = 7$; the SDP-bounded variant certifies gains only away from the incumbent; one 320 nm two-dimensional gap and the large-F random two-dimensional gaps remain unresolved. Near-degenerate instances stress first-order solvers. Literature verification relied on abstracts and search snippets; the full texts of [1, 8, 15, 16, 19] were not accessible, so overlap with the body of those works cannot be excluded.

Data and Code Availability

The repository https://github.com/oleksiizadorozhnyi/Fable-Research contains the code (`code/`), every result file (`results/`) and the verification record. For every verified instance `results/certs/<name>.npz` archives the sample coordinates with their bright/dark labels, the order set, the source points and weights, the focus set and index, the dual multipliers (μ, λ), the scale κ and the bound; `results/exp*_rbest.npy` the best masks, `results/exp*_Y.npy` the lifted solutions (multi-exposure ensembles), `results/exp*_dual.json` the duals, and `results/environment.txt` the Python, numpy, scipy, cvxpy, CLARABEL, SCS and mpmath versions. Random instances use fixed seeds (42 for the random one-dimensional set, 7 for the expanded dataset, 0 for the nonnegative-mask search). `python3 -W ignore test_certlitho.py` runs the thirteen unit tests; `./run_all_par.sh` regenerates every result file; `exp21_exact_verify.py` re-verifies every certificate; `make_numbers.py` regenerates every number and table body of this manuscript.

Acknowledgements and Competing Interests

This manuscript was prepared by an autonomous research system under human oversight; six independent adversarial reviews (mathematics, code, data consistency, lithographic physics, prior art, editing) were applied before this version. No external funding was received. The authors declare no competing interests.

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